

Analytical MCQ Assessment

High-level cognitive evaluation paper.

SECTION A: Multiple Choice Questions

Q1. When testing a mobile application's behavior under poor network conditions, which approach provides the most realistic simulation of latency and packet loss on a physical device? (1 Marks)

- A. Using a mock API response with delayed timers.
- B. Utilizing a network conditioning tool like Charles Proxy or Link Conditioner.
- C. Disabling Wi-Fi and relying solely on the local emulator's network settings.
- D. Reducing the CPU speed of the test device.

Q2. During mobile app testing, a tester switches the app to the background, opens several resource-intensive apps, and then returns to the original app. What specific behavior is being evaluated? (1 Marks)

- A. Memory leak detection.
- B. State restoration and resource reclamation.
- C. Interruption handling.
- D. Boundary value analysis.

Q3. What is a primary architectural advantage of using Espresso or XCUITest over Appium for mobile test automation? (1 Marks)

- A. They support cross-platform execution using a single codebase.
- B. They run inside the application's process, allowing faster execution and synchronization with the UI thread.
- C. They do not require access to the application's source code.
- D. They support a wider range of programming languages.

Q4. A security tester is attempting to intercept HTTPS traffic from a mobile application using a proxy tool but receives SSL handshake errors. What security mechanism is likely causing this? (1 Marks)

- A. Root detection.
- B. Code obfuscation.
- C. SSL Pinning.
- D. Two-Factor Authentication.

Q5. Which of the following is the most effective metric to monitor when investigating why a mobile application causes excessive battery drain? (1 Marks)

- A. Total installation package size (APK/IPA).
- B. CPU wake locks and background network activity.
- C. Frame rate (FPS) during idle states.
- D. Memory allocation in the heap.

Q6. What is the primary difference between an Alpha test and a Beta test of a mobile application? (1 Marks)

- A. Alpha testing is done by the development team, while Beta testing is done by external users.
- B. Alpha testing uses emulators, while Beta testing is performed only on physical devices.
- C. Alpha testing focuses on security, while Beta testing focuses on performance.
- D. Alpha testing is automated, while Beta testing is entirely manual.

Q7. When automating a mobile application, which W3C Actions API capability is essential for simulating a pinch-to-zoom gesture? (1 Marks)

- A. A single pointer performing a rapid double-tap.
- B. Two pointers moving simultaneously in opposite directions.
- C. A single pointer performing a long press followed by a swipe.
- D. Two pointers moving sequentially in the same direction.

Q8. How should a mobile QA team best address the challenge of device fragmentation with a limited budget? (1 Marks)

- A. Test exclusively on the latest flagship device of each operating system.
- B. Use a cloud-based device farm to run automated tests on a representative matrix of target devices.
- C. Rely solely on emulators and simulators for all phases of testing.
- D. Postpone device testing until users report bugs in production.

Q9. What does 'Upgrade Testing' specifically verify in a mobile application lifecycle? (1 Marks)

- A. The app's ability to run on a newer version of the mobile operating system.
- B. The preservation of user data and application state when installing a new app version over an older one.
- C. The performance of the app when the device's hardware is upgraded.
- D. The speed of downloading the app from the App Store or Google Play.

Q10. When analyzing a crash report from an iOS application, what process is required to convert memory addresses into human-readable function names and line numbers? (1 Marks)

- A. Decompilation.
- B. Symbolication.
- C. Obfuscation.
- D. De-serialization.

INSTRUCTOR ONLY - ANSWER KEY & RUBRIC

MCQ Solutions & Rationale

Q1. Correct Answer: B

Rationale: Network conditioning tools intercept and throttle actual network traffic to realistically simulate latency and packet loss.

Q2. Correct Answer: B

Rationale: This test verifies if the app correctly restores its state after the OS reclaims its memory while in the background.

Q3. Correct Answer: B

Rationale: Native frameworks run inside the app's process, enabling automatic synchronization with the UI thread and faster execution.

Q4. Correct Answer: C

Rationale: SSL Pinning restricts the app to accepting only specific, predefined certificates, blocking proxy-intercepted traffic.

Q5. Correct Answer: B

Rationale: Keeping the CPU awake via wake locks and frequent background network requests are primary drivers of battery consumption.

Q6. Correct Answer: A

Rationale: Alpha testing is typically conducted internally by developers/testers, whereas Beta testing involves a wider audience of external users.

Q7. Correct Answer: B

Rationale: Pinch-to-zoom requires multi-touch simulation where two pointers move towards or away from each other simultaneously.

Q8. Correct Answer: B

Rationale: Cloud device farms provide cost-effective access to a diverse matrix of real physical devices for automated testing.

Q9. Correct Answer: B

Rationale: Upgrade testing ensures that local databases, user preferences, and session states are preserved during an app update.

Q10. Correct Answer: B

Rationale: Symbolication replaces memory addresses in crash logs with readable source code method names and line numbers using dSYM files.